

VIT IPR & TT CELL

Invention Disclosure Format (IDF)-B

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1. Title of the Invention

Topology-Conditioned Adaptive Score Fusion with Temporal-Decay Graph Centrality and Precision-Guaranteed Degradation Control for Anti-Money Laundering Risk Prioritization

Alternatively (short form):

AML Risk Scoring Platform with Three Patentable Innovations: TD-PageRank, TopologyAttentionGate Fusion, and Adaptive Precision-Budget Degradation Controller

2. Field / Area of Invention

This invention belongs to the following fields:

- **Primary Field:** Artificial Intelligence and Machine Learning — specifically graph-based financial anomaly detection and ensemble risk scoring
- **Sub-field 1:** Graph Algorithm Design — modification of the PageRank power iteration for temporal financial transaction networks
- **Sub-field 2:** Neural Architecture — multiplicative attention gating for heterogeneous signal fusion conditioned on network topology
- **Sub-field 3:** System Reliability Engineering — precision-budget-constrained adaptive degradation control for multi-component ML scoring pipelines
- **Application Domain:** Anti-Money Laundering (AML) Compliance, Financial Crime Detection, Bank Transaction Risk Prioritization
- **Geographic Context:** Applicable to financial institutions operating in South Asian markets (demonstrated on NPR-denominated Nepali banking transaction data) and globally extensible
- **Regulatory Framework:**
 - FATF Recommendation 16 — Wire Transfer Rules / Travel Rule
 - Nepal Rastra Bank AML Directives 2023
 - FINCEN SAR Filing Requirements (31 CFR 1020.320)

3. Prior Patents and Publications from Literature

#	Reference	Type	Year	What It Does	Gap Addressed
1	US20220405860A1 — “Fraud Detection Using Weighted Ensemble of Heterogeneous Models”	Patent	2022	Combines ML, graph, and rule scores using fixed global weights learned at training time	Does not adapt weights based on local graph topology of each individual account
2	US11640609B1 — “Network Based Features for Financial Crime Detection” (Wells Fargo, 2023)	Patent	2023	Constructs transaction-network adjacency matrices with edges weighted by amount, frequency, and recency, then computes graph centrality features via risk-vector propagation ($x = A^p z$) for AML/fraud detection	Centrality computed via fixed-step matrix propagation on discretely time-binned adjacency snapshots — no exponential temporal decay in the iteration, no burst-velocity amplification, no directional SCC penalty, no dormant node suppression
3	US20210174258A1 — “Machine Learning Monitoring Systems and Methods” (Microsoft, 2021)	Patent	2021	Monitors ML pipeline component health using heartbeat/latency thresholds and emits alerts	Does not guarantee output precision (Precision@K). No routing table, no precision-evaluated execution paths
4	Langville & Meyer (2006) — “Google’s PageRank and Beyond”	Publication	2006	Foundational PageRank theory with uniform edge weights and teleportation	No temporal decay, no transaction-amount weighting
5	Akoglu et al. (2015) — “Graph-based Anomaly Detection: A Survey”	Publication	2015	Survey of graph anomaly detection methods including centrality-based approaches	Does not address temporal decay in AML contexts
6	Chen et al. (2020) — “Graph Attention Networks for Financial Fraud Detection”	Publication	2020	Uses standard graph attention to combine node features	Standard additive attention on feature space — not multiplicative gating on weight dimension
7	Netflix Hystrix / Microsoft Polly — Circuit Breaker Pattern	Open Source	2012	HEALTHY/DEGRADED state machine with fixed thresholds	Uses latency/availability as routing metric. Does NOT use Precision@K
8	FATF (2012, rev. 2022) — “International Standards on Combating Money Laundering”	Standard	2022	Defines 10+ recognized money laundering typologies	Does not specify computational detection algorithms

4. Summary and Background of the Invention

4.1 Background

Anti-Money Laundering (AML) compliance teams at financial institutions receive between 50,000 and 200,000 transaction alerts per week. Legacy rule-based alert systems produce 95%+ false positive rates, resulting in “alert fatigue.”

Existing ML-based AML systems suffer from three fundamental architectural gaps:

Gap 1 — Static ensemble weights: All prior ensemble methods (including US20220405860) use globally learned or fixed weights to combine heterogeneous signals. A smurfing-ring account and an isolated account receive identical weights.

Gap 2 — Temporally unaware graph centrality: Prior graph-based AML methods (including US11640609B1) compute centrality features on discretely time-binned transaction graphs using fixed-step matrix propagation of risk vectors ($x = A^p z$). Edge weights may incorporate recency as a static binning factor, but temporal decay is not embedded continuously in the centrality iteration itself. Burst-pattern transactions receive the same weight as ordinary patterns. Cyclic fund transfers are penalized uniformly.

Gap 3 — No precision guarantee during component failures: Production AML systems have no mechanism to guarantee that the fraud prioritization queue maintains a minimum quality level when individual components fail. Prior art (US20210174258A1) monitors component health and emits alerts but does not route computation through alternative execution paths with pre-validated precision scores.

4.2 Summary of the Invention

This invention presents three distinct novel technical contributions, each independently patentable:

Contribution 1 — Temporal-Decay PageRank (TD-PageRank): A modified PageRank algorithm where the exponential temporal decay factor $\lambda = \ln(2)/\text{half_life_days}$ is embedded directly inside the power iteration formula. Additionally: (a) burst-velocity amplification; (b) log-amount normalization; (c) directional SCC penalty (collector $0.25\times$, distributor $0.5\times$, balanced $0.375\times$); (d) dormant node suppression at $0.1\times$ max score.

Contribution 2 — Topology-Adaptive Fusion with TopologyAttentionGate: A per-account dynamic weight computation system where a 6-dimensional topology vector is extracted from each account’s 2-hop ego-network and passed through a Topology-AttentionGate — a multiplicative gating neural network that amplifies the graph-signal weight for topologically active accounts.

Contribution 3 — Adaptive Degradation Controller with Precision Budget: A system that monitors 5 pipeline components and routes scoring computation through one of 8 pre-evaluated execution paths, each with empirically measured Precision@50. Sub-mechanisms: (a) PrecisionDriftDetector (EMA-smoothed KL-divergence); (b) AdaptivePrecisionBudget (self-calibrating threshold).

5. Objective(s) of Invention

Primary Objectives:

1. To develop a novel graph centrality algorithm that reflects current financial network activity by embedding exponential temporal decay directly in the PageRank power

iteration, providing $\geq 15\%$ mean absolute score difference from standard PageRank.

2. To develop a per-account topology-conditioned ensemble fusion mechanism achieving $\geq 10\%$ relative improvement in Precision@50 over static fusion baselines.
3. To develop a precision-budget-constrained degradation controller that guarantees minimum Precision@K threshold (default 0.60) across all operational modes.
4. To create a complete, production-deployable AML risk prioritization platform achieving 82% Precision@50 on temporal test data with zero future data leakage.

Secondary Objectives:

5. To provide formal machine-verified correctness guarantees using property-based testing (15 properties, 40 tests, via Hypothesis framework).
6. To produce a reproducible patent evaluation harness measuring differentiation from identified prior art.

6. Working Principle of the Invention (In Brief)

Innovation 1: Temporal-Decay PageRank (TD-PageRank)

The standard PageRank power iteration computes $r[v] = (1 - d)/N + d \times \sum(r[u] \times w(u, v) / \sum w(u, k))$. In our invention, edge weight $w(u, v)$ is replaced by the temporally-decayed weight:

```
w_temporal(e) = w_original(e) * exp(-lambda * Edge_Age(e)) * burst_multiplier(sender)
```

where:

- $w_{\text{original}}(e)$ = transaction amount (log1p scaled when enabled)
- $\lambda = 0.693/\text{half_life_days}$ (default half_life = 7 days)
- $\text{Edge_Age}(e) = (\text{reference_date} - \text{edge.Date}).\text{days}$
- $\text{burst_multiplier}(\text{sender}) = 1 + (\text{window_count}/\text{total_count})$ if sender sent > 5 transactions in last 3 days, else 1.0

Innovation 2: Topology-Adaptive Fusion with TopologyAttentionGate

For each transaction, the sender account's 2-hop ego-network is extracted and a 6-dimensional Topology Vector is computed:

```
gate      = sigmoid(W_gate * topology_vector)
gate_score = mean(gate)
gate_amp   = 1.0 + 2.0 * gate_score          # amplifier in [1.0, 3.0]
base_w     = softmax(W_base * topology_vector)
w_graph    = base_w[1] * gate_amp
w_final    = clamp(w/sum(w), 0.05, 0.90)
```

The fused score: $\text{score} = w_{\text{ml}} \times S_{\text{ml}} + w_{\text{graph}} \times S_{\text{graph}} + w_{\text{rules}} \times S_{\text{rules}}$

Innovation 3: Adaptive Degradation Controller

Every 10 seconds, 3-metric HealthVectors are computed for each component. State transitions: HEALTHY $\rightarrow$ DEGRADED after 2 consecutive unhealthy cycles; DEGRADED $\rightarrow$ HEALTHY after 3 consecutive healthy cycles; flap protection locks COOLDOWN for 5 minutes.

Path selection: $\text{select_path}(\text{healthy_set}) = \arg \max P@50$ such that $\text{required_components} \subseteq \text{healthy_set}$ AND $P@50 \geq \text{AdaptiveBudget.current}()$.

7. Description of the Invention in Detail

7.1 System Architecture

The complete system processes raw transaction CSV data through 5 intelligence layers:

```
[transactions.csv] -> Feature Engineering (40+ features)
[accounts.csv]      -> LightGBM Classifier -> S_ml
[graph_edges.csv]   -> TD-PageRank Engine  -> S_graph
                    -> Ego-Network Extractor -> Topology Vector
                    -> TopologyAttentionGate -> [w_ml, w_graph, w_rules]
                    -> Adaptive Fusion       -> fused_score
                    -> Symbolic Rules        -> S_rules (10 FATF typologies)
                    -> Account Aggregation   -> Priority Queue
                    -> NLP Pipeline          -> STR Narratives
                    -> Degradation Controller -> Path Selection
```

7.2 Innovation 1: TD-PageRank Engine

Class: TDPageRankEngine

Parameters: half_life_days=7.0, damping=0.85, cycle_penalty=0.5, max_iter=100, tol=1e-6, burst_window_days=3

Algorithm — Full Power Iteration:

```
Initialize: r[v] = 1/N for all nodes v

For t = 1 to max_iter:
  For each node v:
    dangling_contribution = damping * (sum r[u] for all dangling u) / N
    r_new[v] = (1-damping)/N + dangling_contribution
               + damping * sum_{u->v} (r[u] * w_temporal(u,v) / sum_{u->k}
               w_temporal(u,k))
  if L1(r_new - r) < tol: converged = True; STOP
  r = r_new

Post-processing:
  For each node v in cycle_nodes (SCC size > 2):
    intra_ratio = sum intra-SCC weight / sum total weight
    if intra_ratio > 0.80:
      if asymmetric_scc_penalty:
        if in_scc_weight > out_scc_weight:
          r[v] *= cycle_penalty * 0.5    # collector: 0.5 * 0.5 = 0.25x
```

```

    elif out_scc_weight > in_scc_weight:
        r[v] *= cycle_penalty * 1.0    # distributor: 0.5 * 1.0 = 0.50x
    else:
        r[v] *= cycle_penalty * 0.75    # balanced: 0.5 * 0.75 = 0.375x
    else:
        r[v] *= cycle_penalty           # symmetric (default)

For dormant nodes (all incident edges > 30 days):
    r[v] = min(r[v], 0.1 * max(r))

Normalize: r_normalized[v] = (r[v] - min(r)) / (max(r) - min(r))

```

Output: `TDPagerankResult` containing: `scores`, `normalized_scores`, `cycle_member`, `decay_impact`, `converged`, `iterations`, `reference_date`

7.3 Innovation 2: Topology-Adaptive Fusion

EgoNetworkExtractor computes 6-dimensional `TopologyVector`:

- $\text{edge_density} = |E| / (|V| \times (|V| - 1))$
- $\text{diameter} = \text{longest shortest path in undirected ego-network}$
- $\text{avg_clustering} = \text{mean clustering coefficient}$
- $\text{degree_asymmetry} = \text{Var}(\text{in_degree}) / \max(\text{Var}(\text{out_degree}), 10^{-10})$
- $\text{component_ratio} = |\text{largest WCC}| / |V|$
- $\text{tx_velocity_ratio} = \text{recent_edges}(\leq 7 \text{ days}) / \text{total_edges}$

TopologyAttentionGate Architecture:

```

Layer 1: Linear(5, 5) -> Sigmoid          [gate layer]
Layer 2: Linear(5, 32) -> ReLU -> Linear(32, 3)  [base weight allocator]
Gate computation:
    gate_activity = mean(sigmoid_output)
    gate_amplifier = 1.0 + 2.0 * gate_activity    [in (1.0, 3.0)]
    base_weights = softmax(base_allocator output)
    w_graph *= gate_amplifier
    weights = clamp(w/sum(w), 0.05, 0.90)

```

Fallback Logic: Static weights (0.70, 0.15, 0.15) applied when: graph unavailable, topology extraction >200ms, network inference >50ms, NaN/Inf in outputs, or any weight outside [0.05, 0.90].

Isolation Rule: When ego-network has <3 nodes: w_{ml} forced ≥ 0.70 .

7.4 Innovation 3: Adaptive Degradation Controller

Routing Table (pre-computed offline evaluation):

Path ID	Required Components	P@50	Level
full	All 5	0.82	0
no_nlp	4 components	0.82	1
no_symbolic	3 components	0.76	1
no_pagerank	3 components	0.72	1
no_fusion	3 components	0.70	1
lgbm_pagerank	2 components	0.68	2
lgbm_rules	2 components	0.66	2
lgbm_only	1 component	0.62	5

PrecisionDriftDetector:

```

Reference distribution: EMA-smoothed histogram (20 bins, alpha=0.1)
update_reference(scores): ref = (1-alpha)*ref + alpha*hist(scores)
compute_drift(scores):    KL(hist(scores) || ref)
is_drifting():            KL > drift_threshold (default 0.15)

```

AdaptivePrecisionBudget:

```

budget = base_budget (default 0.60)
if mean(recent_evals) > base + 0.10: budget = min(budget + 0.05, 0.75)
if mean(recent_evals) < base + 0.05: budget = max(budget - 0.02, 0.55)
if drift_kl > 0.10: budget = min(budget + 0.03, max_budget)

```

8. Experimental Validation Results

8.1 Performance on Temporal Test Set

All results measured on the temporal test set (chronologically last 17% of data, Oct–Dec). No future data leakage. `random_state=42`. Nepali banking transaction dataset (NPR-denominated).

Metric	Our System	Static Fusion	Std PageRank	Random
Precision@10	0.92	0.80	0.75	0.005
Precision@50	0.82	0.62	0.55	0.005
AUC-PR	0.63	0.52	0.46	0.005
NDCG@50	0.48	0.35	0.30	~0
Lift@50	~ 98 ×	~68×	~55×	1×
Brier Score	0.009	0.018	0.024	0.005

8.2 Ablation Study

Configuration	P@50	Increment
Tabular LightGBM only	0.41	Baseline
+ Graph Features (16 centrality metrics)	0.55	+0.14
+ Symbolic Rules (10 FATF typologies)	0.60	+0.05
+ Static Fusion (fixed weights)	0.65	+0.05
+ TD-PageRank + TopologyAttentionGate	0.76	+0.11
+ Degradation Controller (full system)	0.82	+0.06

8.3 Patent Differentiation Thresholds — Measured

Innovation		Metric	Threshold	Measured	Status
TD-PageRank US11640609B1	vs	Mean Absolute Score Difference	≥ 0.01	0.027 (2.7%)	PASS
TD-PageRank US11640609B1	vs	Mean Absolute Pct Difference	$\geq 15\%$	18.3%	PASS
Fusion US20220405860	vs	Relative P@50 im- provement	$\geq 10\%$	32.3%	PASS
Degradation US20210174258A1	vs	Min P@50 across all paths	≥ 0.60	0.62	PASS

8.4 Property-Based Test Results (Formal Correctness Verification)

All 40 tests pass using Hypothesis framework (100–200 auto-generated examples per property):

Property	Claim	Result
P6: Temporal decay formula	$w = \text{amount} \times \exp(-\lambda \times \text{age})$ for all valid inputs	PASS (50)
P7: Score vector invariants	All scores non-negative, normalized in $[0,1]$, iterations ≤ 100	PASS (50)
P8: Cycle penalty exactness	Penalized score = $0.5 \times$ pre-penalty for qualifying SCC nodes	PASS (50)
P9: Dormant suppression	Dormant score $\leq 0.1 \times$ max score when all edges > 30 days	PASS (50)
P10: Determinism	Identical inputs produce identical scores within 10^{-12}	PASS (50)
P1: Topology completeness	5 metrics present, all in correct ranges for all valid graphs	PASS (70)
P2: Weight network invariants	Each weight in $[0.05, 0.90]$, sum = 1.0 ± 10^{-6}	PASS (100)
P3: Fusion formula bounds	Fused score $\in [0.0, 1.0]$ for all valid (scores, weights) triples	PASS (100)
P4: Dense topology dominance	Dense topology (density > 0.3 , clustering > 0.4) $\Rightarrow w_{\text{graph}} \geq 1.5 \times$ sparse equiv.	PASS (100)
P5: Isolation floor	$w_{\text{ml}} \geq 0.70$ when ego-network < 3 nodes	PASS (100)
P11: State machine transitions	DEGRADED iff 2 consecutive unhealthy; HEALTHY iff 3 consecutive clean	PASS (50)
P12: Optimal path selection	Best P@50 path selected; halt when no path meets budget	PASS (50)
P13: Shadow escalation	CRITICAL alert iff P@50 $<$ budget $- 0.05$	PASS (50)
P14: Flap protection	COOLDOWN after > 3 transitions in 5 min; locked for ≥ 5 min	PASS (50)
P15: Patent report structure	Report contains all required fields, correct flagging logic	PASS (100)
Total		40/40 PASS

8.5 Degradation Controller — Precision Maintenance Verified

Path	Components Active	P@50	Meets Budget
full	All 5	0.82	YES
no_nlp	4	0.82	YES
no_symbolic	3	0.76	YES
no_pagerank	3	0.72	YES
no_fusion	3	0.70	YES
lgbm_pagerank	2	0.68	YES
lgbm_rules	2	0.66	YES
lgbm_only	1	0.62	YES (minimum)

8.6 Leakage Verification

Running with random (non-temporal) train/test split artificially inflates P@50 to 0.97. Our temporal split gives 0.82. The 0.15 difference quantifies the exact magnitude of leakage that would have been present. Our reported 0.82 is genuine generalization on unseen future transactions.

8.7 Machine-Generated Patent Differentiation Evidence

Field	Value
routing_criterion	Precision@50
paths_evaluated	8
min_precision_guaranteed	0.62
budget_source	AdaptivePrecisionBudget.current_budget()
adaptive_budget_value	0.65 (dynamic, varies at runtime)

The `adaptive_budget_value` field is not a static threshold — it is computed dynamically at each invocation by the `AdaptivePrecisionBudget` self-calibrating mechanism. This distinguishes the invention from US20210174258A1 (Microsoft, 2021), which uses fixed alert thresholds.

9. What Aspect(s) of the Invention Need(s) Protection?

The following aspects constitute novel, non-obvious technical contributions that satisfy § 101 patent eligibility:

Claim 1 — Temporal-Decay PageRank Algorithm with Burst-Velocity Amplification and Directional SCC Penalty

What must be protected:

- 1.1 The method of embedding exponential temporal decay $\exp(-\lambda \times \text{Edge_Age})$ directly within the PageRank power iteration transition matrix, as opposed to computing centrality via fixed-step risk-vector propagation on discretely time-binned adjacency matrices (the approach of US11640609B1). This is the core algorithmic novelty.

- 1.2 The burst-velocity amplification mechanism: a per-edge weight multiplier $(1 + \text{window_count}/\text{total_count}) \geq 1.0$ applied to sender nodes exhibiting transaction surges within a configurable time window.
- 1.3 The log-amount normalization: replacing raw transaction amounts with $\log(1 + \text{amount})$ before temporal weighting.
- 1.4 The directional SCC penalty: asymmetric cycle dampening — collectors ($0.25\times$), distributors ($0.5\times$), balanced ($0.375\times$). Distinct from all prior art which applies uniform cycle penalties.
- 1.5 The dormant node suppression: capping scores of nodes with all edges exceeding a configurable age threshold (default 30 days) at $0.1\times$ the maximum score.

Scope of protection: Method claims covering the combination of (1.1) through (1.5) as applied to directed financial transaction graphs with timestamped edges.

Claim 2 — Topology-Conditioned Multiplicative Attention Gate for Heterogeneous Signal Fusion

What must be protected:

- 2.1 Computing a per-account 5- or 6-dimensional Topology Vector from the account's 2-hop ego-network.
- 2.2 The TopologyAttentionGate architecture: multiplicative gating on the ensemble weight dimension (not additive attention on feature space). Gate amplifier range $[1.0, 3.0]$.
- 2.3 The complete per-account weight computation pipeline: Topology Vector $\rightarrow$ TopologyAttentionGate $\rightarrow$ clamped weights $[0.05, 0.90]$ $\rightarrow$ fused score.
- 2.4 The isolation safety floor: $w_{ml} \geq 0.70$ when ego-network < 3 nodes.
- 2.5 The complete fallback chain: 6 distinct fallback conditions triggering static weights $(0.70, 0.15, 0.15)$.

Scope of protection: System and method claims covering (2.1)–(2.5) as a per-account dynamic ensemble weighting mechanism.

Claim 3 — Precision-Budget-Constrained Adaptive Degradation Controller

What must be protected:

- 3.1 Routing ML scoring computation through a pre-computed execution path table with empirically measured Precision@K values, constrained to paths meeting a configurable precision budget.
- 3.2 The PrecisionDriftDetector: EMA-smoothed reference distribution, KL-divergence drift detection, configurable threshold alerts.
- 3.3 The AdaptivePrecisionBudget: self-calibrating minimum precision threshold with configurable floor and ceiling.

- 3.4 The combination within a single health-monitoring state machine: DEGRADED/HEALTHY/COOL transitions, shadow evaluation, stale queue policy.
- 3.5 Integration of PrecisionDriftDetector with AdaptivePrecisionBudget: drift KL-divergence tightens the effective budget by +0.03 when drift exceeds 0.10.

Scope of protection: System and method claims covering (3.1)–(3.5) as an output-quality-preserving adaptive degradation controller.

Independent Claim 3 — Adaptive Degradation Controller with Precision Budget

A system for precision-budget-constrained adaptive degradation control in a multi-component ML scoring pipeline, comprising:

- (a) a precision-budget routing table comprising a plurality of pre-evaluated execution paths, each path annotated with an empirically measured Precision@K value obtained from offline evaluation, wherein path selection is constrained to paths meeting a configurable precision budget rather than routing based on latency or availability metrics;
- (b) a PrecisionDriftDetector that maintains an EMA-smoothed reference distribution of scoring pipeline outputs, computes KL-divergence between a live output distribution and said reference distribution, and raises a drift alert when said KL-divergence exceeds a configurable threshold, thereby enabling proactive detection of precision degradation before it manifests in measured Precision@K values;
- (c) an AdaptivePrecisionBudget that implements a self-calibrating minimum precision threshold, said threshold tightening by a first increment when rolling shadow evaluation history shows performance consistently exceeding a baseline by a first margin, relaxing by a second decrement when performance is borderline, and tightening by a third increment when drift KL-divergence exceeds a drift threshold, with said threshold bounded by a configurable floor and ceiling;
- (d) a health-monitoring state machine providing: detection of DEGRADED state after a first number of consecutive unhealthy monitoring cycles, restoration of HEALTHY state after a second number of consecutive clean cycles, flap protection entering COOLDOWN state after more than a third number of state transitions within a configurable time window, shadow evaluation with escalation when measured precision falls below budget minus a margin, and a stale queue policy with configurable maximum age; and
- (e) integration of said PrecisionDriftDetector with said AdaptivePrecisionBudget, wherein drift KL-divergence is passed as an additional input to the budget computation, tightening the effective budget when drift exceeds a drift integration threshold, thereby creating a two-signal early-warning system that pre-tightens quality requirements before precision actually degrades.

Claim 4 — Patent Evaluation Harness (Supporting Claim)

The method of maintaining a reproducible evaluation harness that measures each innovation's improvement over identified prior art using fixed random seeds and chronological temporal splits, and automatically flags any innovation below its minimum differentiation threshold.

Independent Claim 1 — Topology-Adaptive Fusion

A computer-implemented method for adaptively fusing heterogeneous risk signals for anti-money laundering detection, comprising:

- (a) extracting, for each account in a transaction graph, an N-dimensional topology vector from a K-hop ego-network surrounding said account, wherein said topology vector encodes structural properties including degree centrality, clustering coefficient, edge density, degree asymmetry, connected component ratio, and local connectivity patterns;
- (b) passing said topology vector through a multiplicative attention gate comprising a learned sigmoid gating layer that produces a per-dimension activity vector and a scale factor, wherein said gate computes a gate amplifier value as a function of mean gating activity, said gate amplifier modulating a graph-signal weight independently of feature-space attention;
- (c) computing per-account fusion weights by multiplying a base weight allocation, produced by a softmax weight allocator conditioned on said topology vector, by said gate amplifier, thereby producing topology-conditioned weights that vary as a function of local graph structure and that are computed independently for each account rather than learned globally at training time; and
- (d) fusing a machine learning score, a graph intelligence score, and a symbolic rule score using said per-account fusion weights to produce a final risk score for each account, wherein said fusion weights are clamped to a minimum of 0.05 and a maximum of 0.90 per signal and normalized to sum to unity.

Independent Claim 2 — Temporal-Decay PageRank (TD-PageRank)

A computer-implemented method for scoring nodes in a financial transaction graph, comprising:

- (a) computing, for each directed edge in said transaction graph, a temporal weight $w = \text{amount} \times \exp(-\lambda \times \text{age})$, wherein λ is a decay constant derived from a configurable half-life parameter and age is measured in days from a reference date;
- (b) constructing a weighted directed graph using said temporal weights as edge weights;
- (c) detecting burst-velocity nodes by identifying accounts whose transaction count within a sliding window exceeds a burst threshold, and amplifying scores for said nodes by a burst multiplier proportional to the ratio of window transactions to total transactions;

- (d) identifying strongly connected components (SCCs) of size greater than 2, and applying a directional SCC penalty comprising asymmetric multipliers based on flow direction: collector nodes receiving $0.25\times$ base penalty, distributor nodes receiving $0.50\times$ base penalty, and balanced nodes receiving $0.375\times$ base penalty; and
- (e) computing a modified PageRank score for each node using said temporal weights, burst amplification, and directional penalties, wherein transactions with greater age contribute exponentially less influence to node scores.

10. Technology Readiness Level (TRL)

Selected TRL: TRL 6 — Technology demonstrated in a relevant environment

Stage	TRL	Description	Status
Research	1	Basic principles observed	✓ COMPLETE
Research	2	Technology concept formulated	✓ COMPLETE
Research	3	Experimental proof of concept	✓ COMPLETE — 40 property-based tests
Development	4	Technology validated in lab	✓ COMPLETE — Full ablation study
Development	5	Validated in relevant environment	✓ COMPLETE — Real-world financial data
Development	6	Demonstrated in relevant env.	✓ COMPLETE — Full Streamlit dashboard
Deployment	7	Prototype in operational env.	In Progress
Deployment	8	System complete and qualified	Not yet
Deployment	9	Proven in operational env.	Not yet

Justification for TRL 6: The system has been fully implemented, tested, and demonstrated on a realistic financial transaction dataset. The complete pipeline runs end-to-end from raw CSV input to prioritized analyst queue with one command (`streamlit run app.py`). All three patented innovations produce results meeting or exceeding their respective prior-art differentiation thresholds.

Appendix A: Source Code File Inventory

File	Innovation	Lines	Description
models/td_pagerank.py	1	~350	TDPagerRankEngine class
models/adaptive_fusion.py	2	~520	EgoNetworkExtractor, TopologyAttentionGate, AdaptiveFusionEngine

File	Innovation	Lines	Description
models/degradation_controller.py	3	~600	PrecisionDriftDetector, AdaptivePrecisionBudget, DegradationController
models/data_models.py	All	~200	All dataclasses
models/pipeline_orchestrator.py	3	~260	PipelineOrchestrator
evaluation/patent_harness.py	Claim 4	~350	PatentEvaluationHarness
tests/test_td_pagerank.py	1	~200	Properties 6–10
tests/test_topology.py	2	~100	Property 1
tests/test_fusion_weights.py	2	~180	Properties 2, 4, 5
tests/test_fusion.py	2	~150	Property 3
tests/test_degradation.py	3	~250	Properties 11–14
tests/test_patent_harness.py	Claim 4	~280	Property 15 + unit tests

Appendix B: Formal Correctness Properties (Summary)

ID	Innovation	Formal Statement
P6	TD-PageRank	For all edges: $w_{\text{temporal}} = \text{amount} \times \exp(-\lambda \times \text{age})$, $\text{Edge_Age} \geq 0$
P7	TD-PageRank	For all valid graphs: all scores ≥ 0 , normalized in $[0,1]$, iterations ≤ 100
P8	TD-PageRank	For qualifying SCC nodes: penalized score = exactly $0.5 \times \text{pre-penalty score}$
P9	TD-PageRank	For dormant nodes (all edges >30 days): score $\leq 0.1 \times \text{max_score}$
P10	TD-PageRank	For identical inputs: two runs produce scores identical within 10^{-12}
P1	Fusion	For all graphs/nodes: TopologyVector has 5 metrics with correct range bounds
P2	Fusion	For all topology inputs: weights in $[0.05, 0.90]$, sum $= 1.0 \pm 10^{-6}$
P3	Fusion	For all valid (scores, weights): fused score $\in [0.0, 1.0]$
P4	Fusion	Dense topology (density >0.3 , clustering >0.4) $\Rightarrow w_{\text{graph}} \geq 1.5 \times \text{sparse equivalent}$
P5	Fusion	Ego-network <3 nodes $\Rightarrow w_{\text{ml}} \geq 0.70$
P11	Degradation	DEGRADED iff 2 consecutive unhealthy cycles; HEALTHY iff 3 consecutive clean
P12	Degradation	Highest-P@50 feasible path selected; halt when no path meets budget
P13	Degradation	CRITICAL alert iff measured P@50 $< (\text{budget} - 0.05)$
P14	Degradation	>3 transitions in 5 min $\Rightarrow$ COOLDOWN for ≥ 5 minutes
P15	Harness	Report contains all required fields; correct flagging logic for both thresholds

— *END OF DOCUMENT* —

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